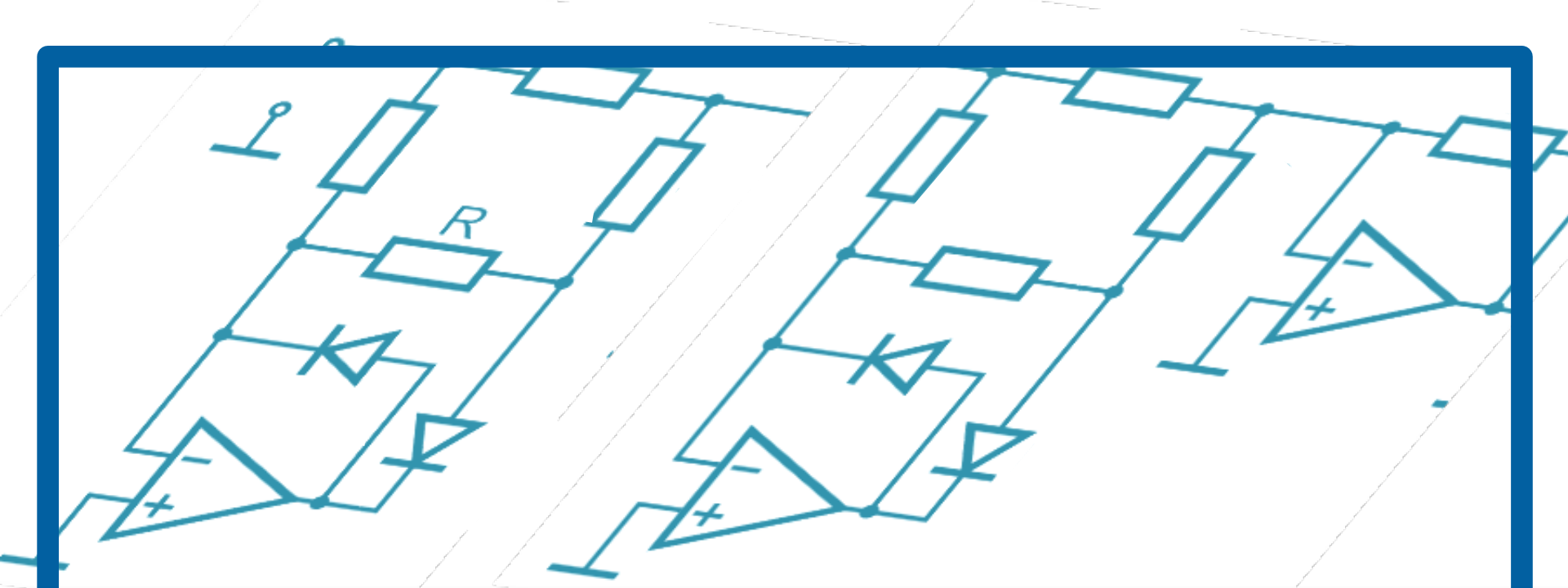




Basic Opamp Circuits

Circuit Analysis & Examples

Learning Objectives

- Study the following commonly used circuits which are designed using opamps:
 - Standard voltage amplifiers
 - ✓ Inverting amplifier
 - ✓ Non-inverting amplifier
 - Summing amplifiers
 - Differential amplifier
 - Voltage follower
 - Current to voltage converter
 - Integrator & differentiator

Opamps in Closed Loop Configuration

- Opamps are not very useful in their open loop configuration as the gain is too high.
- We will now investigate opamp circuits which have their output fed back to the input.
- This is known as the closed loop configuration.
- The closed loop gain G is dependent on the component values of the opamp circuit.
- G is independent of the open loop gain A .

Input & Output Impedance

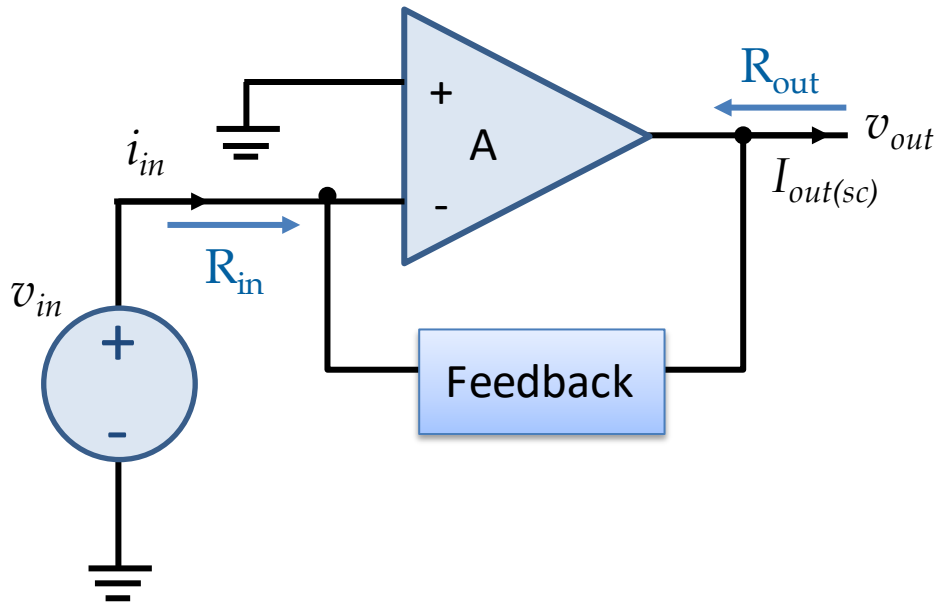
- In closed loop opamp circuits the input and output impedance refer to the impedances (resistance) observed by the input and output voltages.

- $R_{in} = \frac{V_{in}}{i_{in}}$

- This depends on the circuit components.

- $R_{out} = \frac{V_{out}}{i_{out(SC)}}$

Short circuit output current

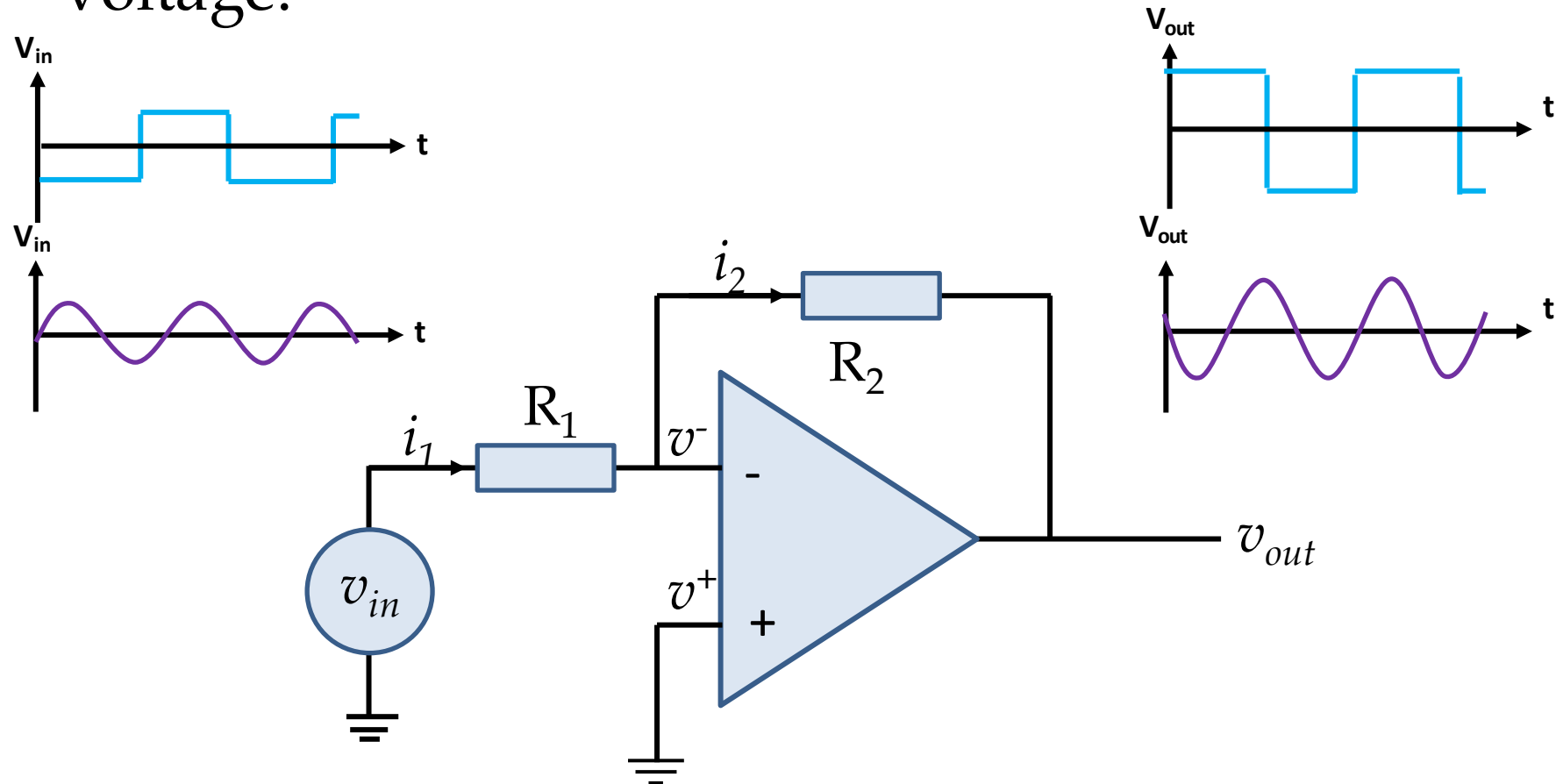


Standard Voltage Amplifiers

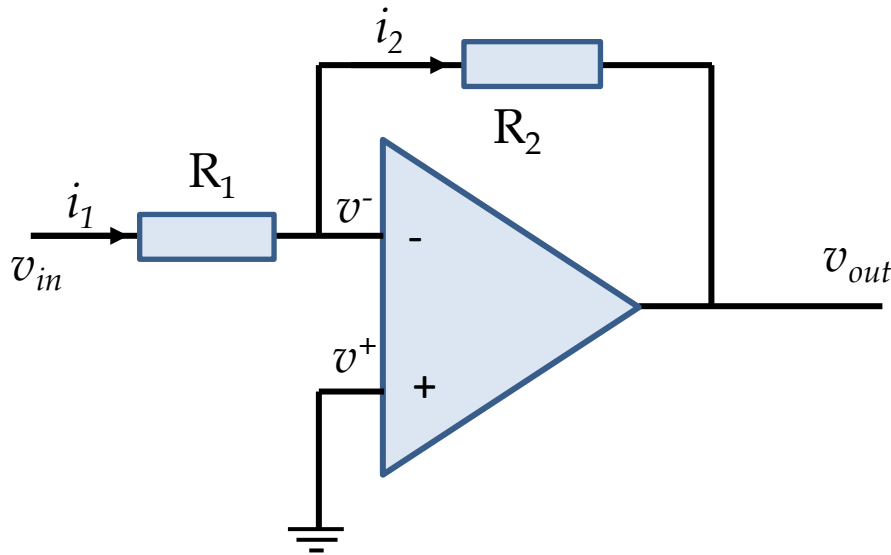
- Two basic configurations:
 - Inverting amplifier: *Input and output are opposite in polarity*
 - Non-inverting amplifier: *Input and output have the same polarity*

Inverting Amplifier

- The output has the opposite polarity to the input voltage.



Inverting Amplifier



$$i_1 = \frac{v_{in} - v^-}{R_1} \quad \& \quad i_2 = \frac{v^- - v_{out}}{R_2}$$

Due to infinite input impedance no current goes into the input terminals of op-amp:

$$\therefore i_1 = i_2$$

$$\frac{v_{in} - v^-}{R_1} = \frac{v^- - v_{out}}{R_2} \quad \text{--- (1)}$$

From virtual earth principle (VEP):

$$v^- = v^+ = 0 \quad \text{--- (2)}$$

Substituting v^- with 0 in (1):

$$\frac{v_{in} - 0}{R_1} = \frac{0 - v_{out}}{R_2}$$

$$v_{out} = -v_{in} \left(\frac{R_2}{R_1} \right)$$

$$G = \frac{v_{out}}{v_{in}} = - \left(\frac{R_2}{R_1} \right)$$

Input & Output Resistances

- The input resistance is:

$$R_{in} = \frac{V_{in}}{i_{in}} = \frac{V_{in}}{i_1} = R_1$$

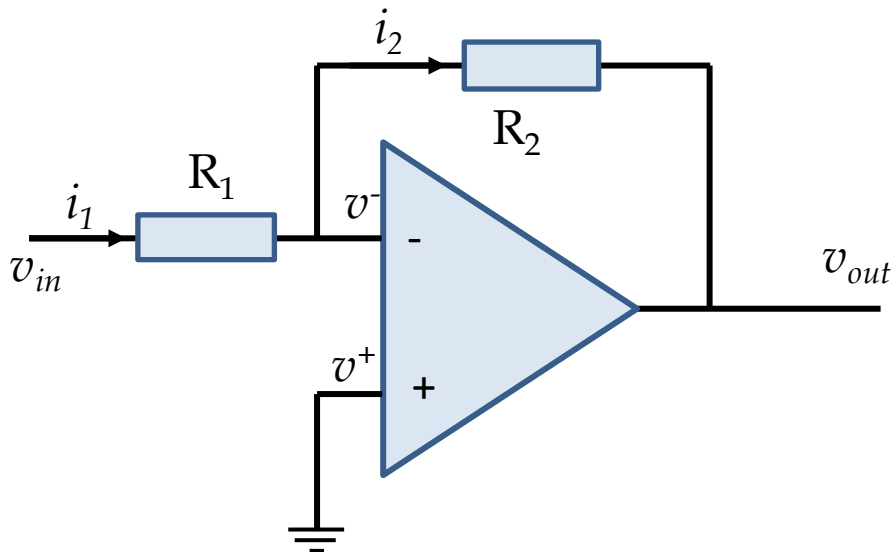
- To avoid signal loss a high input resistance is needed.
- A large resistance must be chosen for R_1 .
- If a large gain is required ($G = R_2/R_1$), R_2 might become impractically large.
- Inverting amplifiers suffer from a low input resistance.
- The output voltage and current are taken at the terminals of the ideal voltage source at output, the output resistance is zero.

Inverting Amplifiers

- It follows the negative feedback.
- The gain factor of these amplifiers is very high.
- The output generated will be out of phase with the applied input signal.
- The potential values at both inverting and non-inverting terminals will be maintained at zero.
- Practical uses include:
 - Phase shifter as the output is 180-degrees out of phase
 - Integration
 - To balance signals
 - Amplifiers for sensors

Inverting Amplifier: Example

Calculate R_1 and R_2 for the given op-amp circuit in order to have a voltage gain of -30 and an input resistance of the circuit to be $10k\Omega$.



$$i_1 = \frac{v_{in} - v^-}{R_1}$$

$$i_2 = \frac{v^- - v_{out}}{R_2}$$

$$i_1 = i_2 \text{ (due to infinite input impedance)}$$

$$\frac{v_{in} - v^-}{R_1} = \frac{v^- - v_{out}}{R_2}$$

$$v_{out} = \frac{(R_1 + R_2) v^- - R_2 v_{in}}{R_1}$$

From the virtual earth principle (VEP):

$$v^- = v^+ = 0$$

$$v_{out} = - \frac{R_2 v_{in}}{R_1}$$

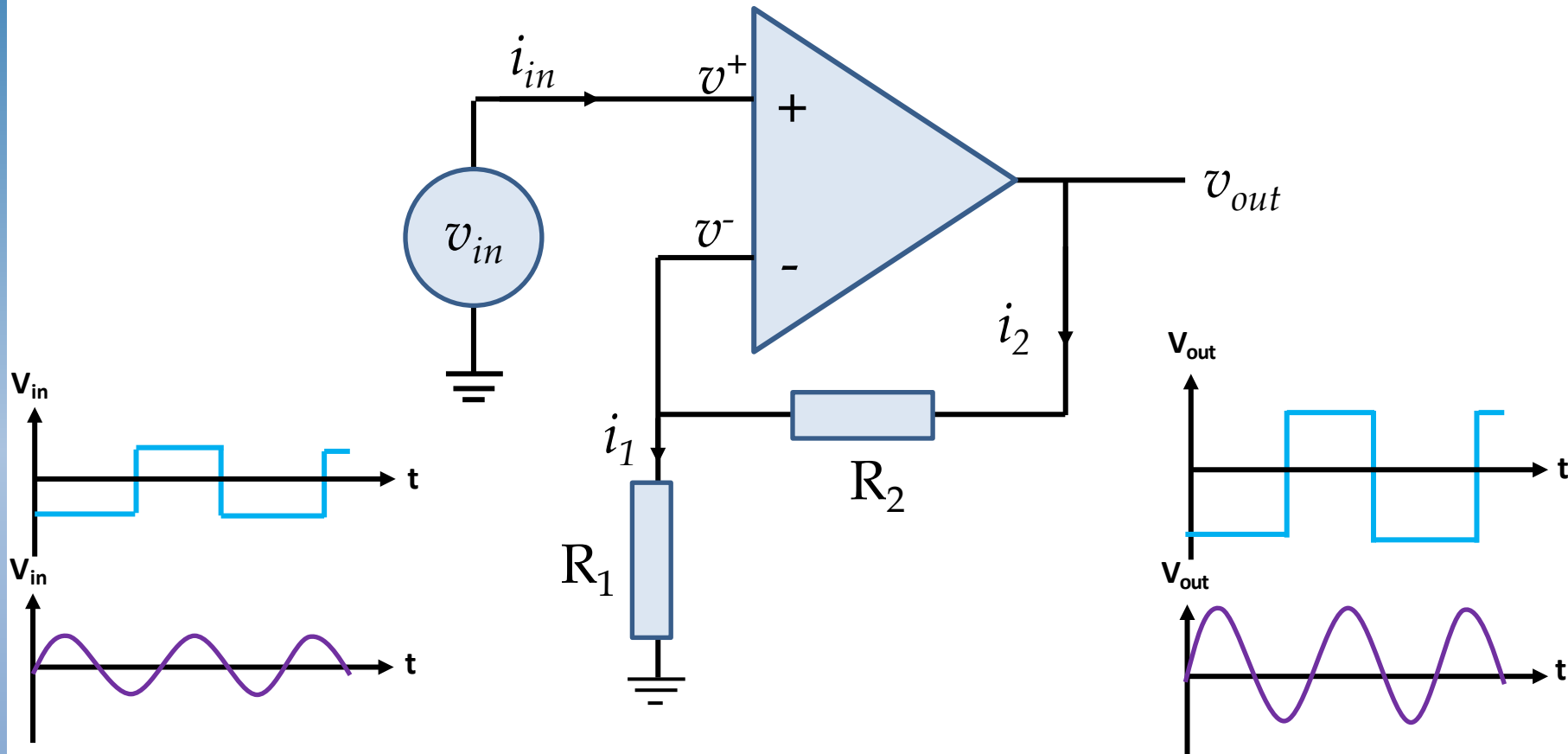
$$R_{in} = \frac{v_{in}}{i_{in}} = \frac{v_{in}}{i_1} = R_1 = 10k\Omega$$

$$\frac{v_{out}}{v_{in}} = - \frac{R_2}{R_1} = -30$$

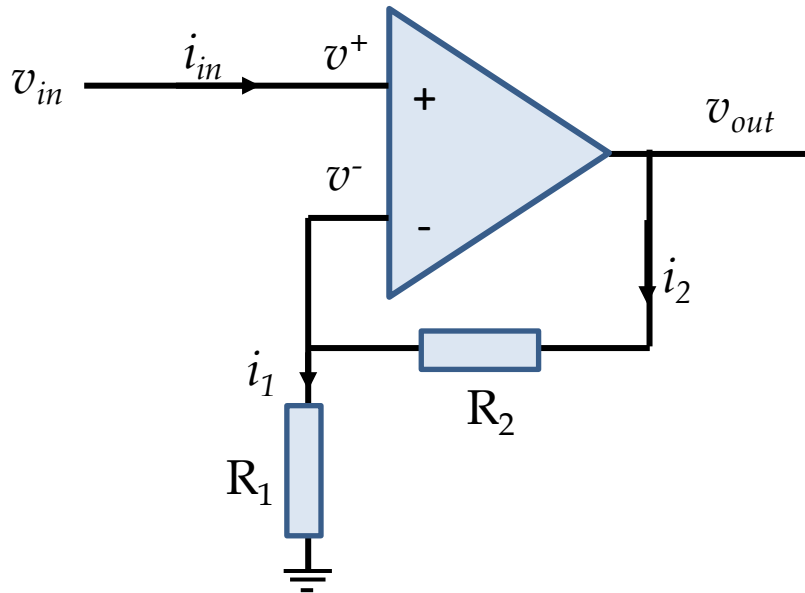
$$R_2 = -(-30) \times 10k\Omega = 300k\Omega$$

Non-inverting Amplifier

- Known as the 'non-inverting amplifier' because the output has the same polarity as input voltage.



Non-inverting Amplifier



$$i_1 = \frac{v^- - 0}{R_1} \quad \& \quad i_2 = \frac{v_{out} - v^-}{R_2}$$

Due to infinite input impedance no current goes into the input terminals of op-amp:

$$\therefore i_{in} = 0$$

$$\text{Also, } i_1 = i_2$$

$$\frac{v^-}{R_1} = \frac{v_{out} - v^-}{R_2} \quad \text{--- (1)}$$

From virtual earth principle (VEP):

$$v^- = v^+ = v_{in} \quad \text{--- (2)}$$

Substituting v^- with 0 in (1):

$$\frac{v_{in}}{R_1} = \frac{v_{out} - v_{in}}{R_2}$$

$$v_{out} = v_{in} \left(\frac{R_1 + R_2}{R_1} \right)$$

$$G = \frac{v_{out}}{v_{in}} = \left(\frac{R_1 + R_2}{R_1} \right)$$

Input & Output Resistances

- The input resistance is:

$$R_{in} = \frac{V_{in}}{i_{in}} = \frac{V_{in}}{0} = \infty$$

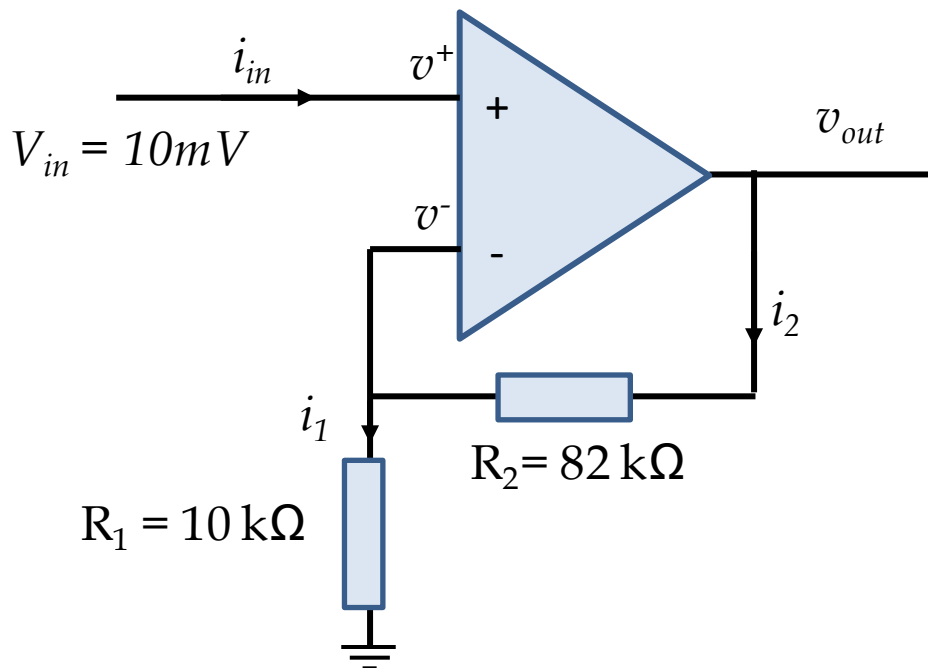
- This is advantageous in some cases.
- The output voltage and current are taken at the terminals of the ideal voltage source at output, the output resistance is zero.

Non-inverting Amplifiers

- The voltage gain is always greater than one.
- The voltage gain is positive, and the output is in-phase with the input signal (for AC) and the output polarity has the same as the input (for DC).
- These are used in
 - Circuits that require a high input impedance
 - To isolate the respective cascaded circuits

Non-inverting Amplifier: Example

In the following op-amp circuit calculate V_{out} , closed loop gain of the op-amp and the input resistance R_{in} .



$$R_{in} = \frac{v_{in}}{i_{in}} \Rightarrow \infty$$

$$\text{as } i_{in} = 0, R_{in} \rightarrow \infty$$

$$i_1 = \frac{v^- - 0}{R_1}$$

$$i_2 = \frac{v_{out} - v^-}{R_2}$$

$$\frac{v^- - 0}{R_1} = \frac{v_{out} - v^-}{R_2}$$

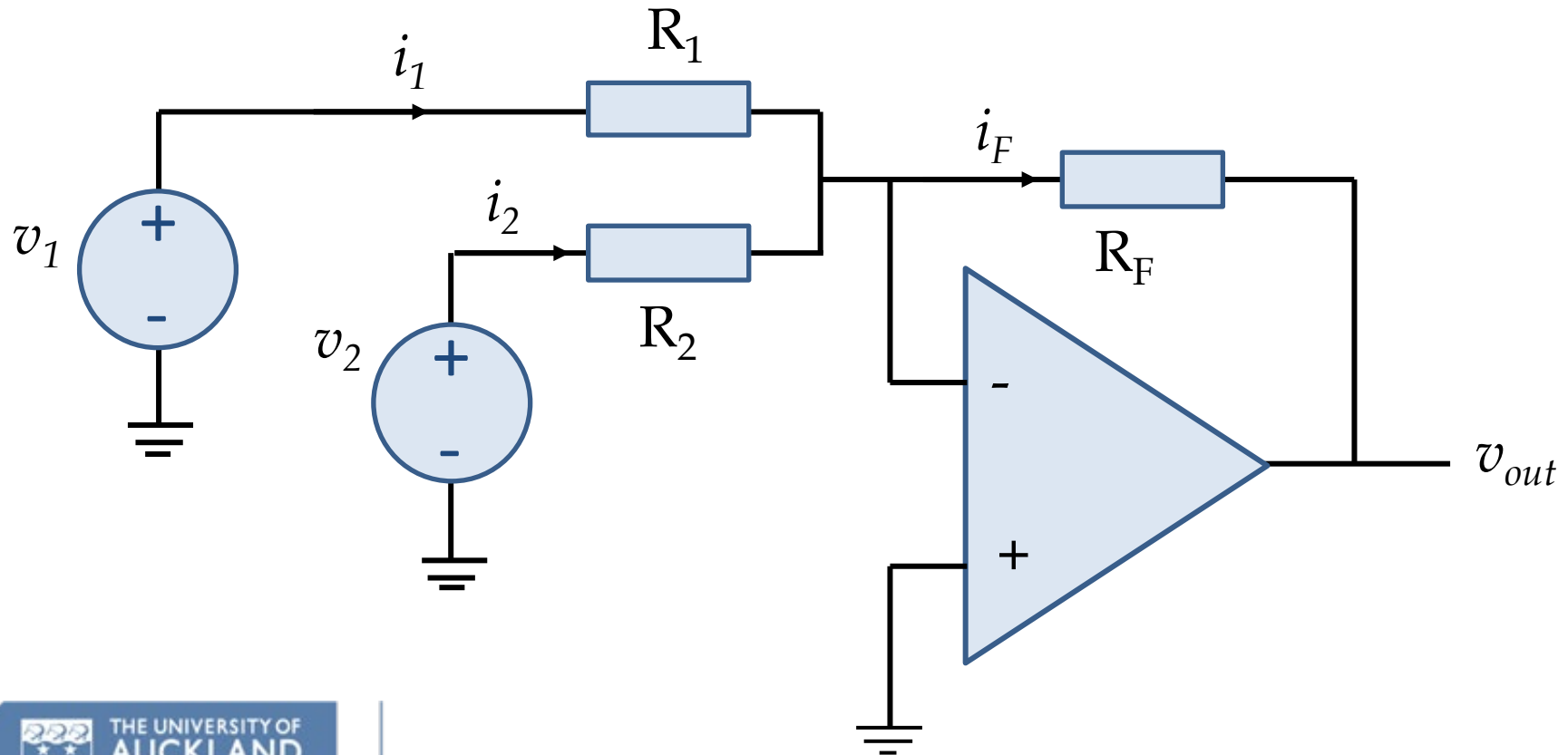
From the virtual earth principle:

$$v^- = v^+ = v_{in}$$

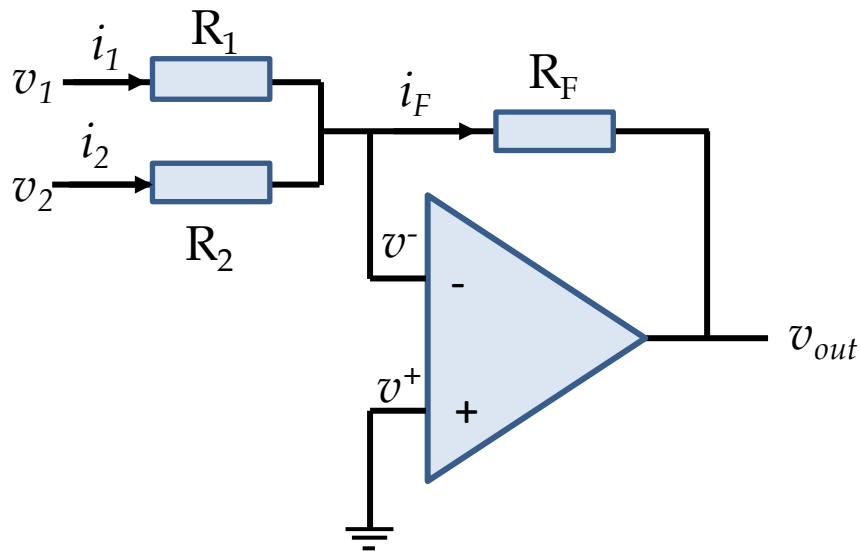
$$\begin{aligned} v_{out} &= \frac{(R_1 + R_2) v_{in}}{R_1} \\ &= \frac{(10k + 82k)}{10k} \times 10mV \\ &= 92 mV \end{aligned}$$

Inverting Summing Amplifier

- Known as the 'inverting summing amplifier' as the output is a weighted sum of the input voltages, which is opposite in polarity.



Inverting Summing Amplifier



$$i_1 = \frac{v_1 - v^-}{R_1} \quad \& \quad i_2 = \frac{v_2 - v^-}{R_2}$$
$$i_F = \frac{v^- - v_{out}}{R_F}$$

From KCL:

$$i_F = i_1 + i_2 \quad \text{--- (1)}$$

Substituting for i_1 , i_2 and i_F in (1):

$$\frac{v^- - v_{out}}{R_F} = \frac{v_1 - v^-}{R_1} + \frac{v_2 - v^-}{R_2}$$

From virtual earth principle (VEP):

$$v^- = v^+ = 0 \quad \text{--- (2)}$$

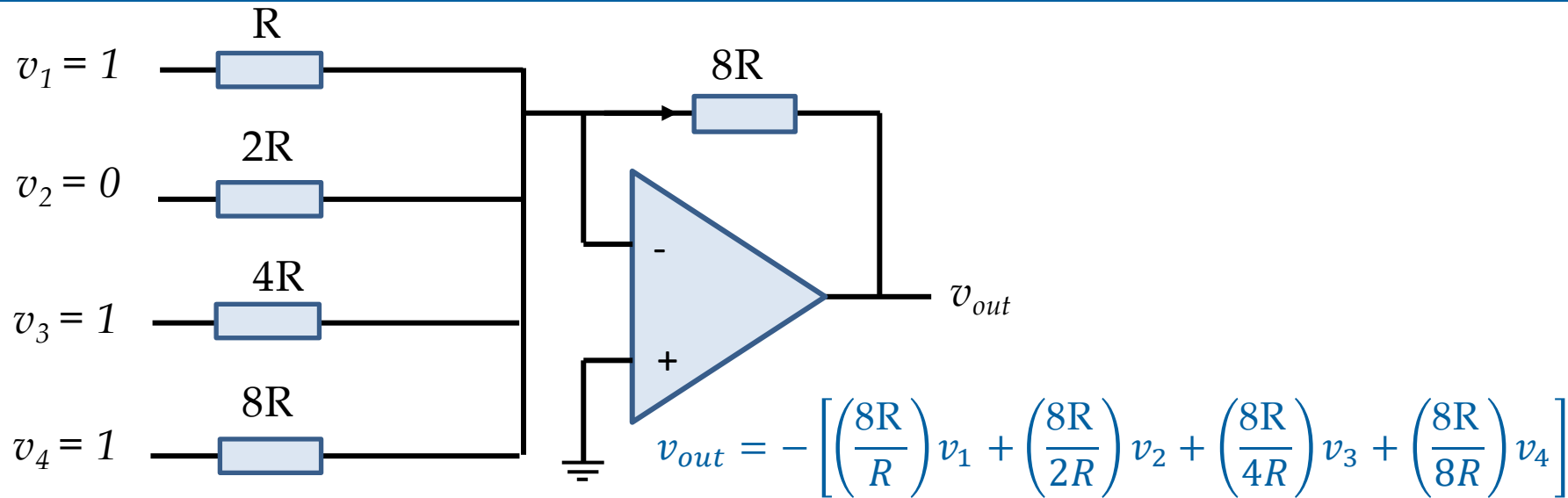
$$\frac{0 - v_{out}}{R_F} = \frac{v_1 - 0}{R_1} + \frac{v_2 - 0}{R_2}$$

$$v_{out} = - \left[\left(\frac{R_F}{R_1} \right) v_1 + \left(\frac{R_F}{R_2} \right) v_2 \right]$$

Summing Amplifiers

- Summing amplifiers can also be designed in the non-inverting configuration.
- These non-inverting summing amplifiers are useful when a phase-shift or inversion of the input signal is undesirable.
- Summing amplifiers are used in:
 - Digital to Analog Converters (DAC)
 - Audio mixers

Digital to Analog Converter (DAC)



- If binary number 1011 is the input to the DAC the output will be:

$$\begin{aligned} v_{out} &= \left| - \left[\left(\frac{8R}{R} \right) 1 + \left(\frac{8R}{2R} \right) 0 + \left(\frac{8R}{4R} \right) 1 + \left(\frac{8R}{8R} \right) 1 \right] \right| \\ &= 8 + 0 + 2 + 1 = 11 \text{ (decimal)} \end{aligned}$$

Audio Mixer

- Summing amplifiers are used for adding multiple signals in audio mixers.

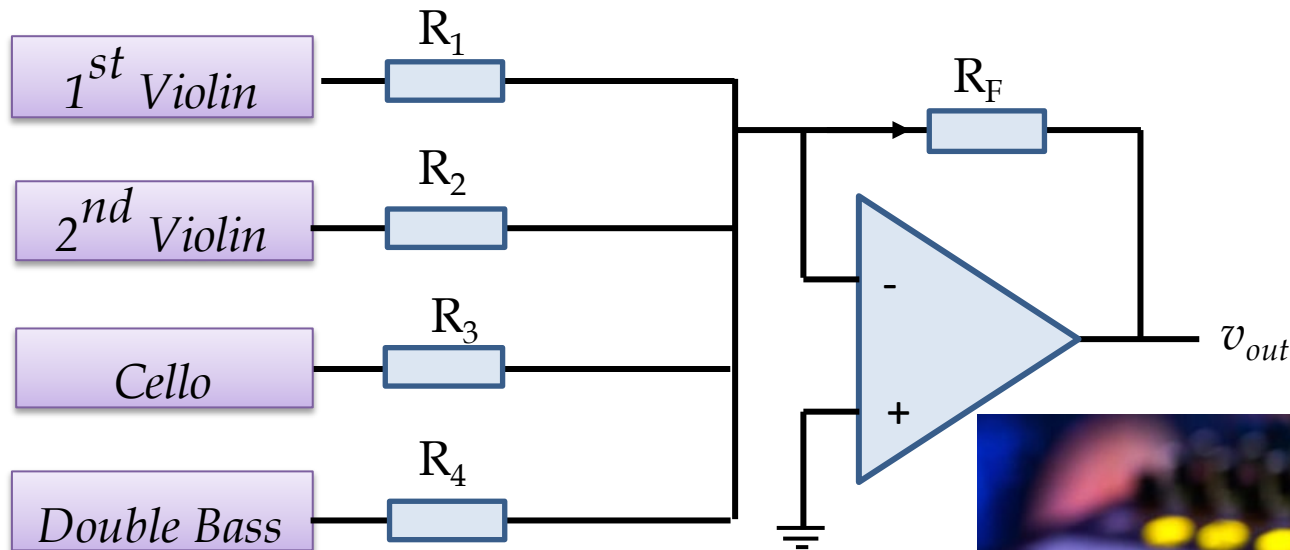
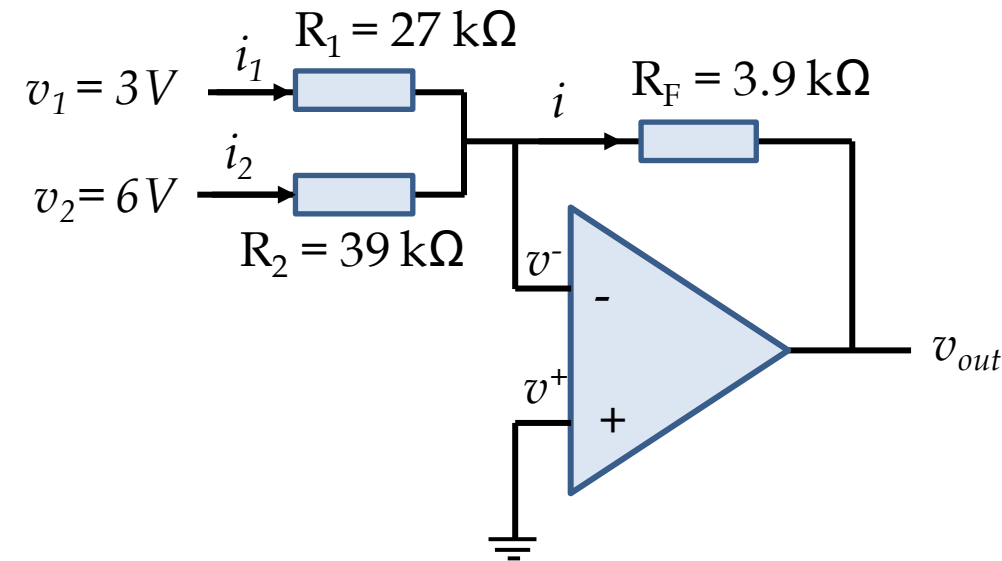


Image source: <https://resources.pcb.cadence.com/blog/2020-summing-amplifier-inversion-properties-and-op-amp-varieties>

Inverting Summing Amplifier: Example

An inverting summing amplifier is shown in the diagram below.

- Calculate the output voltage v_{out} .
- Calculate the current i .



From the virtual earth principle (VEP):

$$v^- = v^+ = 0$$

$$i = \frac{0 - v_{out}}{3.9k}$$

$$\frac{v_1 - 0}{27k} + \frac{v_2 - 0}{39k} = i = \frac{v_{out} - 0}{3.9k}$$

$$\frac{0 - v_{out}}{3.9} = \frac{v_1}{27} + \frac{v_2}{39}$$

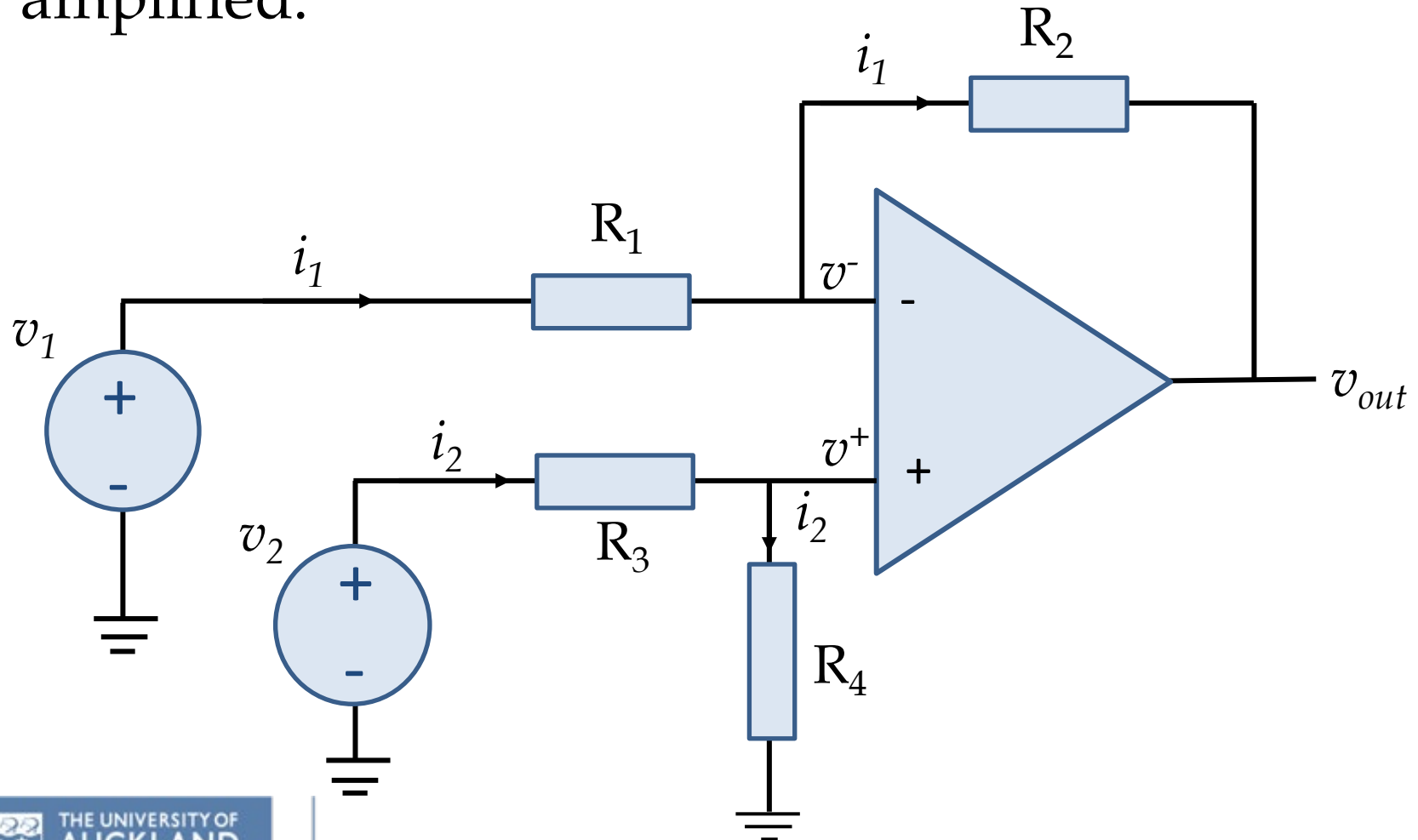
$$v_{out} = -3.9 \times \left(\frac{3}{27} + \frac{6}{39} \right) = -1.033 \text{ V}$$

$$i = -\frac{-1.033}{3.9k}$$

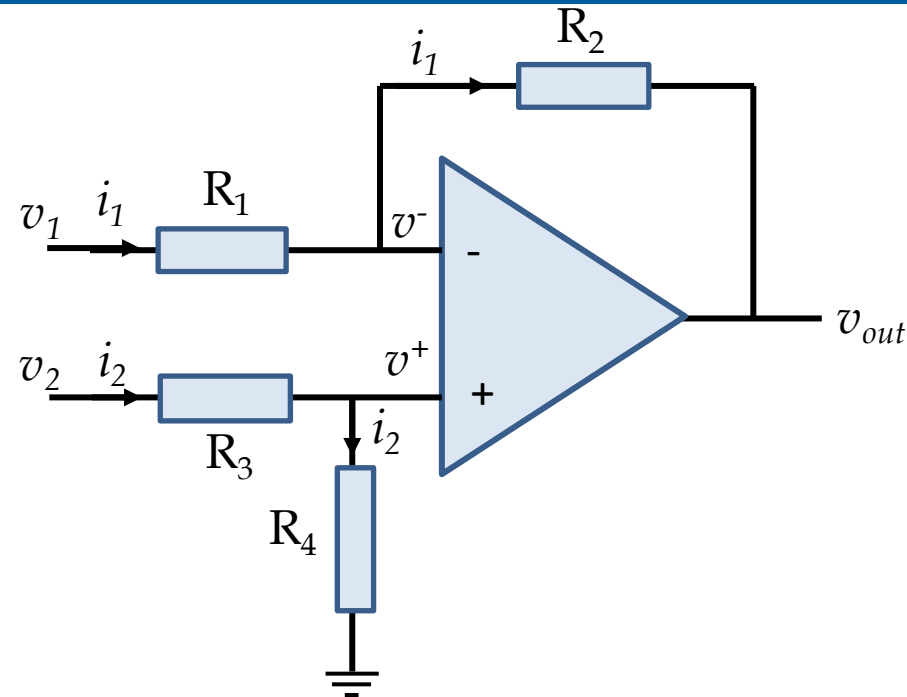
$$= 0.265 \text{ mA}$$

Differential Amplifier

- The difference between the two inputs v_1 & v_2 is amplified.



Differential Amplifier



$$i_1 = \frac{v_1 - v^-}{R_1} = \frac{v^- - v_{out}}{R_2}$$

$$v^- = \frac{R_2 v_1 + R_1 v_{out}}{(R_1 + R_2)} \quad \text{--- (1)}$$

$$i_2 = \frac{v_2 - v^+}{R_3} = \frac{v^+ - 0}{R_4}$$

$$v^+ = \frac{R_4 v_2}{(R_3 + R_4)} \quad \text{--- (2)}$$

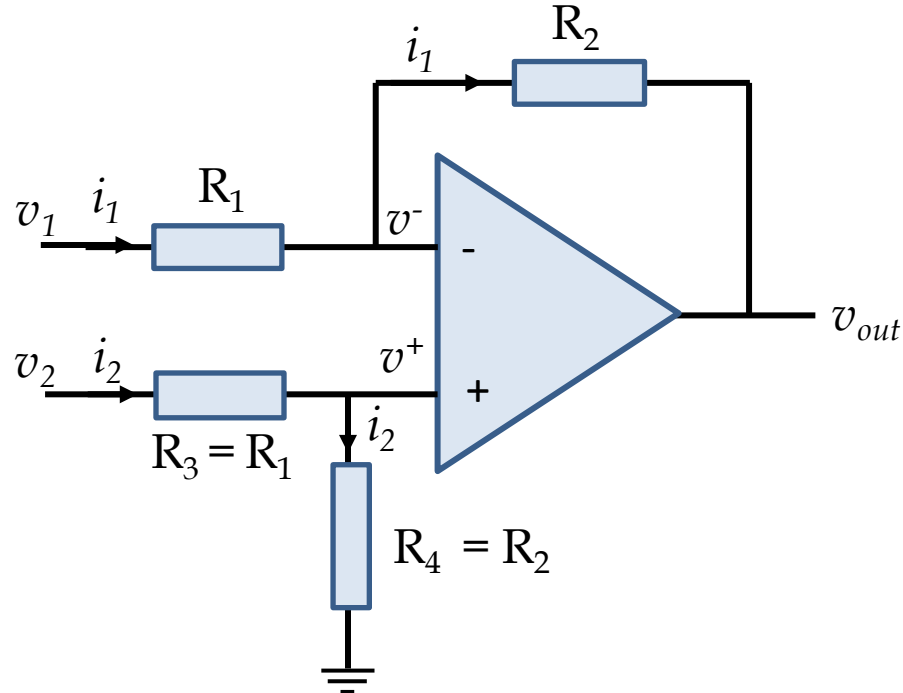
From virtual earth principle (VEP):

$$v^- = v^+ \quad \text{--- (3)}$$

$$\frac{R_2 v_1 + R_1 v_{out}}{(R_1 + R_2)} = \frac{R_4 v_2}{(R_3 + R_4)}$$

$$v_{out} = \frac{R_4 (R_1 + R_2) v_2}{R_1 (R_3 + R_4)} - \frac{R_2 v_1}{R_1}$$

Difference Amplifier



$$v_{out} = \frac{R_4(R_1 + R_2) v_2}{R_1(R_3 + R_4)} - \frac{R_2 v_1}{R_1}$$

$$\text{If } R_1 = R_3 \text{ and } R_2 = R_4: \quad v_{out} = \left(\frac{R_2}{R_1} \right) (v_2 - v_1)$$

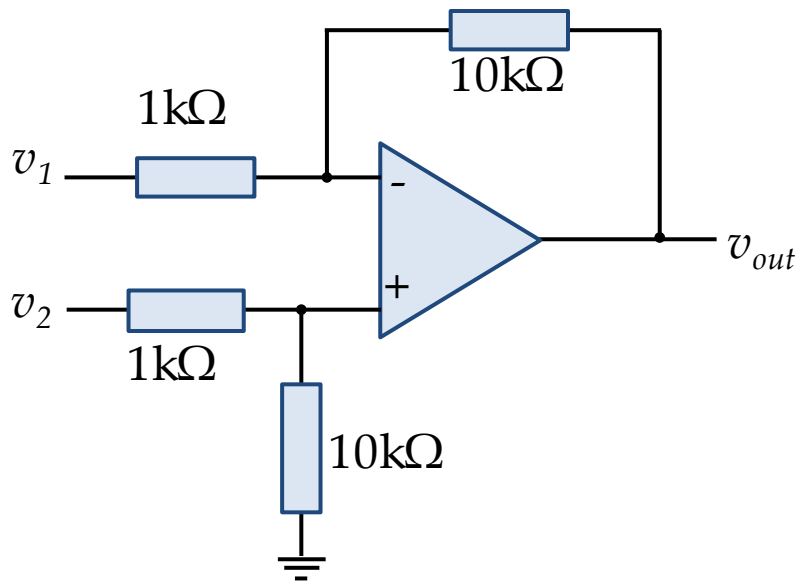
Differential Amplifiers

- Used to amplify the difference between two input voltages while suppressing any voltage common to the two inputs.
- These are used in:
 - Noise suppression
 - Instrumentation amplifiers
 - Motor control circuits
 - Volume control circuits

Differential Amplifier: Example

An operational amplifier circuit is shown below:

- If $v_1 = v_2 = 1V$, what is the output voltage?
- If $v_1 = 1V$ and $v_2 = 0.5V$, what will the output voltage be?



$$v_{out} = \frac{10k}{1k} (v_2 - v_1)$$

$$\text{If } v_1 = v_2 = 1V,$$

$$(v_2 - v_1) = 0V$$

$$v_{out} = 10 (v_2 - v_1) = 0V$$

$$V_{out} = \frac{10k}{1k} (v_2 - v_1)$$

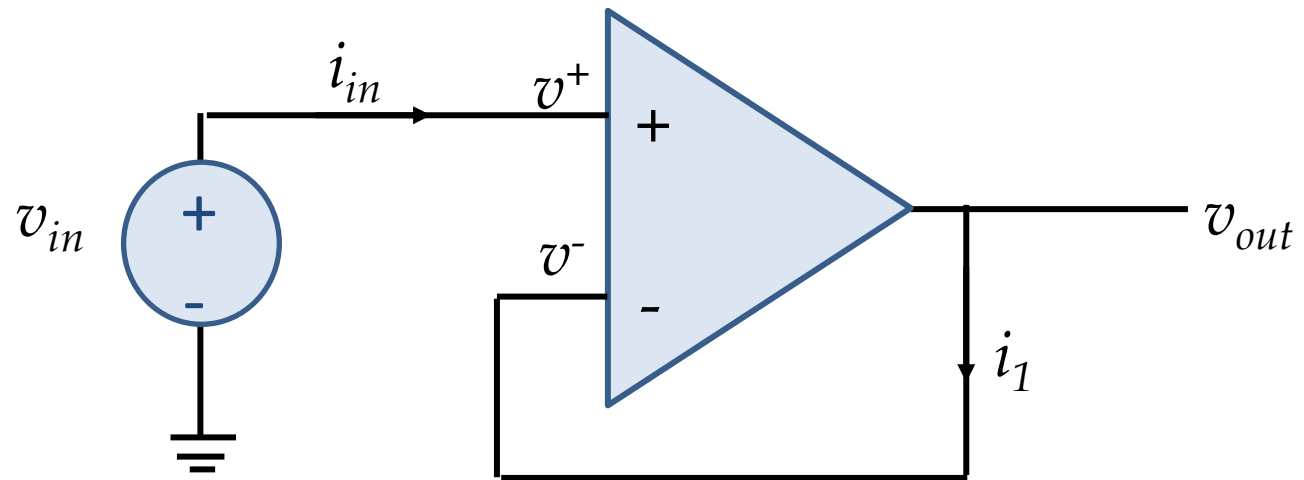
$$\text{If } v_1 = 1V \text{ and } v_2 = 0.5V,$$

$$(v_2 - v_1) = -0.5V$$

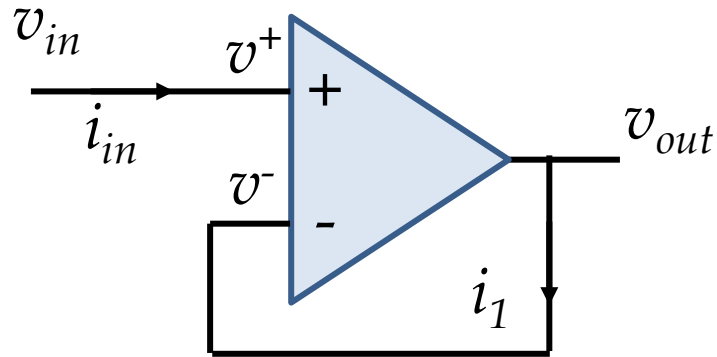
$$v_{out} = 10 (-0.5) = -5V$$

Voltage Follower (Buffer)

- The output *follows* the input but the input signal does not get 'loaded' by the output stage, and to ensure that the output stage does not draw current from the input signal.
- Also referred to as unity-gain amplifier or buffer.
- Used to isolate input and output in a circuit.



Voltage Follower



$$v^+ = v_{in} \quad \text{—————} (1)$$

$$v^- = v_{out} \quad \text{—————} (2)$$

From virtual earth principle (VEP):

$$v^- = v^+$$

$$v_{in} = v_{out}$$

Due to infinite input impedance no current goes into the input terminals of op-amp:

$$i_1 = 0$$

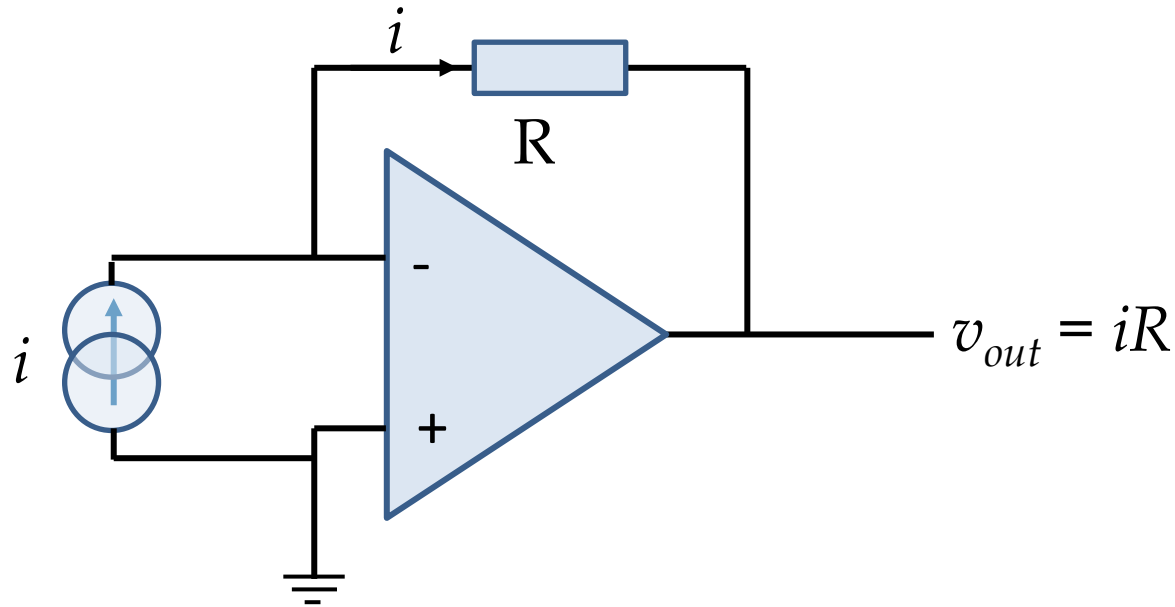
$$i_{in} = 0$$

Input & output resistances:

$$R_{in} = \frac{v_{in}}{i_{in}} = \frac{v_{in}}{0} = \infty$$

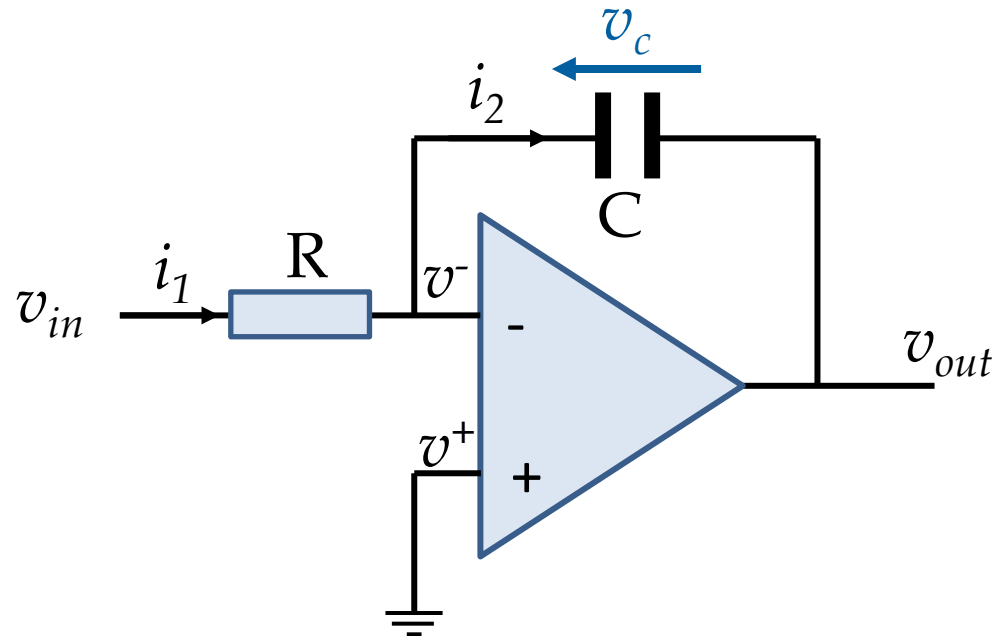
$$R_{out} = 0$$

Current to Voltage Converter



- If the resistance R is fixed v_{out} is dependent on i .
- These are often used in photodetectors where light is turned to a current and then output as a voltage which is directly proportional to the current.

Integrator



$$i_1 = \frac{v_{in} - v^-}{R}$$

From VEP:

$$v^- = v^+ = 0$$

$$i_1 = \frac{v_{in}}{R}$$

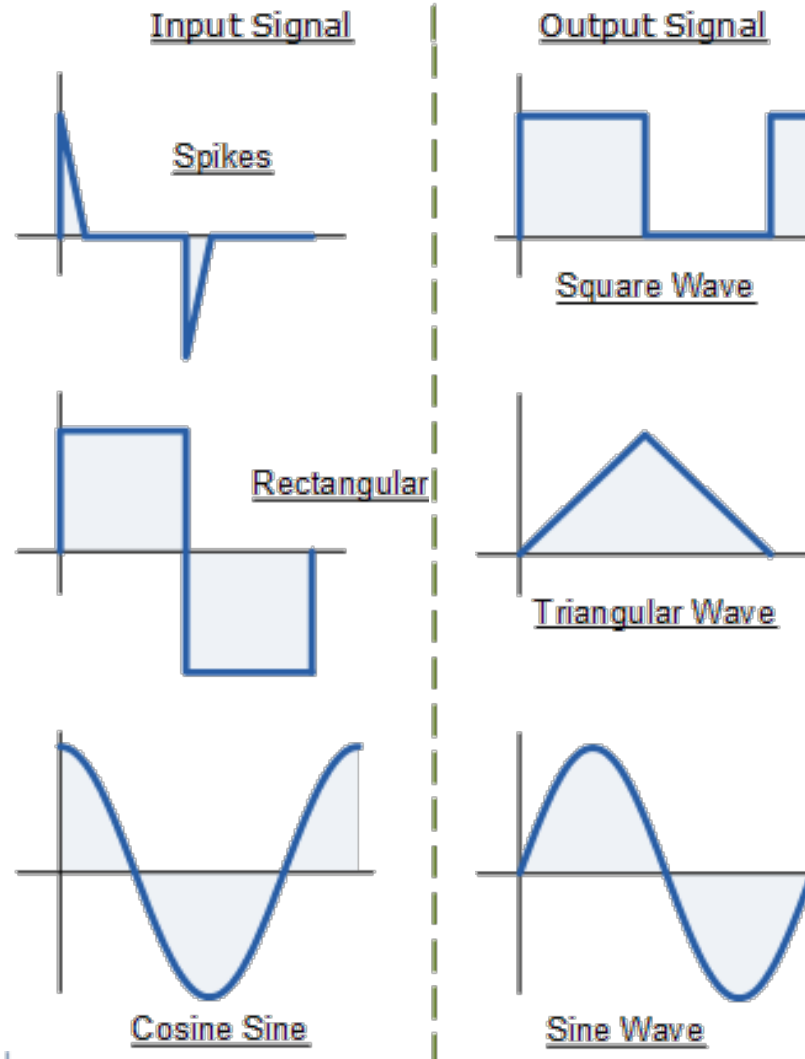
$$i_2 = i_1 = \frac{v_{in}}{R}$$

$$\text{Voltage across capacitor: } v_c = \frac{1}{C} \int i_2 dt = \frac{1}{C} \int \frac{v_{in}}{R} dt = \frac{1}{RC} \int v_{in} dt$$

$$\text{Output voltage: } v_{out} = 0 - v_c = -\frac{1}{RC} \int v_{in} dt$$

Output is the integral of the input voltage

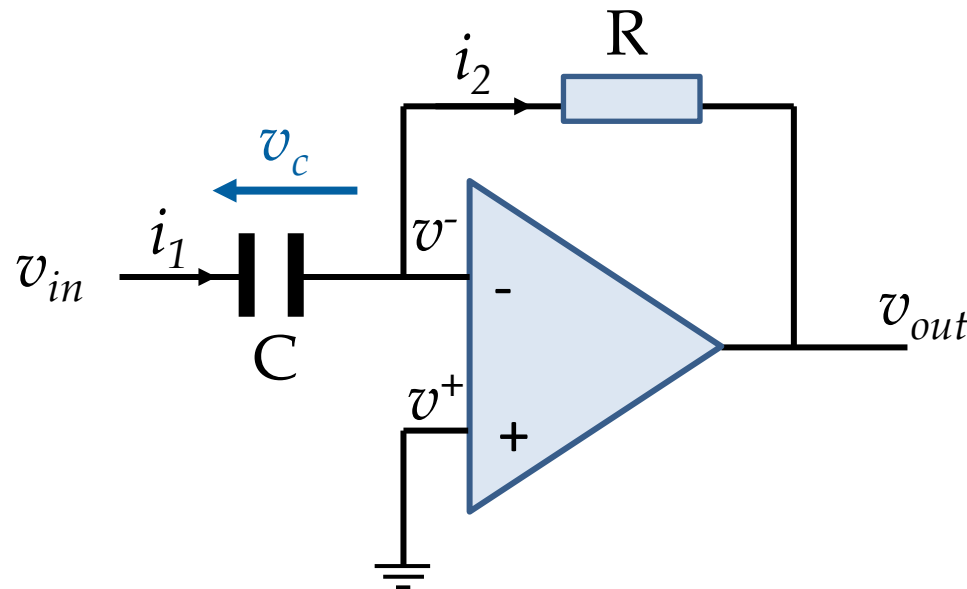
Integrator Waveforms



Integrator

- A square wave can be used to generate a triangular wave using this integrator circuit.
- If the input signal has a low frequency the capacitor may appear as an open circuit and then the opamp will behave as if it was in the open loop configuration.
- To avoid this in real integrators a resistor is connected in parallel with the capacitor.
- Op-amp integrating amplifiers are used in:
 - Calculus operations performers in analogue computers
 - Analogue-to-digital converters
 - Ramp generators
 - Wave shaping applications

Differentiator



$$i_2 = \frac{v^- - v_{out}}{R}$$

From VEP:

$$v^- = v^+ = 0$$

$$i_2 = \frac{-v_{out}}{R}$$

$$v_{out} = -i_2 R$$

Current & voltage through capacitor: $v_c = v_{in} - v^- = v_{in}$

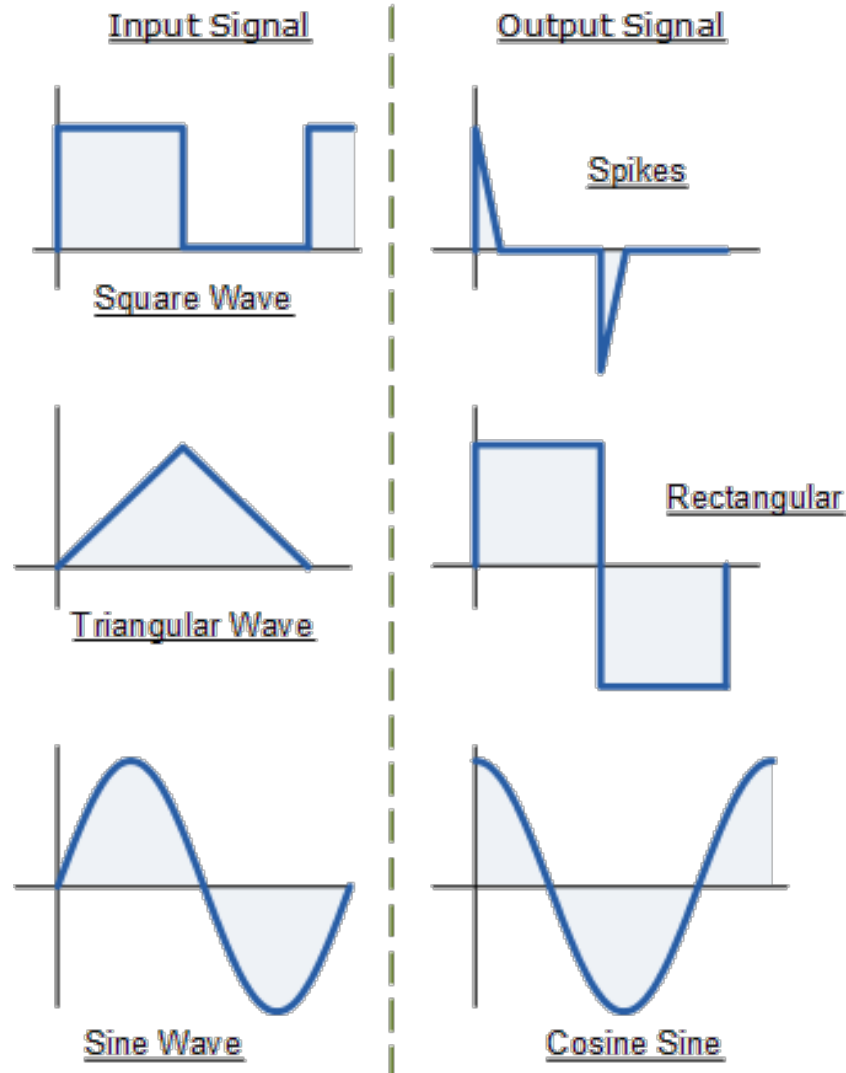
$$i_1 = C \frac{dv_c}{dt} = C \frac{dv_{in}}{dt}$$

Since $i_1 = i_2$, $v_{out} = -i_1 R$

Output voltage: $v_{out} = -i_1 R = -RC \frac{dv_{in}}{dt}$

Output is the differential of the input voltage

Differentiator Waveforms



Differentiator

- One of the most widely used circuits.
- Essentially a high pass filter.
- It can be used as:
 - Square wave generators for a triangle wave input
 - Wave shaping and signal processing circuits
 - Calculation modules in analogue computers
 - PID controllers
 - rate-of-change detectors in frequency modulators